Credit Name: CSE2130 - File Structures & Exception Handling

Assignment Name: Word Guess

Error Log Entry

What error message did you encounter (if any)?

```
Exception in thread "main" java.lang.NullPointerException: Cannot invoke "String.length()" because "secretWord" is null
        at Mastery.WordGuess.main(WordGuess.java:87)
```

What unexpected behavior did your program exhibit?

1. The secret word would not update to the correct word, but would remain empty instead.

2. Despite following the example in the textbook my code still did not correctly update the letters as they were guessed. It only updated the first letter.

What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)

1. Logic Error - Used readFile.readLine() to get the number of words in the file, which read the whole file already, so when I tired to read it again to update the secret word, the file would be empty and return null.

2. Logic Error - The textbook was not the problem, although at first I thought it was, instead the issue was the secret words I put in the file. My code converted all the letters that the user enters into uppercase, so when it compared the letter to the word in the file (that was all lowercase except for the first letter) it wouldn't match.

Include a screenshot of specific lines of code.

1.
```
while(( words = readFile.readLine()) != null)
{
    System.out.println(words);
    numWords += 1;
    System.out.println(numWords);
}

//update the word to guess to the random object and number of words read plus one
wordToGuess = (int)(Math.random() * numWords) + 1;

//iterate through the word to guess slots
for(int i = 0; i < wordToGuess; i++)
{
    //update the secret word from the lines read from the file
    secretWord = readFile.readLine();
}
```

2. No code was the error just the words in my file.

```
Mouse
Planet
Winter
Elephant
Methods
```

```
Guess a Letter ( Enter ! to guess the entire word ):
M
M------

Guess a Letter ( Enter ! to guess the entire word ):
O
M------
```

How did you fix the issue?

1. I changed the method of getting the number of words in the file, which removed the problem of already reading the file to the end.

2. I changed the words in the file to be all uppercase so that it matches the user's guess.

Provide the corrected code or solution using a screenshot.

1.
```
//Get the number of words in the file using readFile
numWords = Integer.parseInt(readFile.readLine());
wordToGuess = ran.nextInt(numWords) + 1;
```

2.
```
5
MOUSE
PLANET
WINTER
ELEPHANT
METHODS
```